

DOCUMENT 2.1 — PROJECT PLANNING

Folder 2: Project Planning Phase | ServiceNow Capstone Project Submission

Project Title: Hospital Appointment Management System
Platform: ServiceNow (Scoped Application)

Application Scope: x_2065601_hospit_0
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1. Executive Summary & Delivery Approach

The **Hospital Appointment Management System** is engineered as a native ServiceNow Scoped Application designed to replace inefficient, error-prone manual outpatient scheduling with a real-time, event-driven digital platform. To ensure structured, high-quality execution within the capstone timeline, an **Agile Development Framework** organized into six structured milestones was adopted.

2. Milestone Schedule & Deliverables

Milestone	Phase / Focus Area	Key Deliverables & Tasks	Status
M1	Ideation & Problem Definition	Problem area identification, 7-feature ideation canvas, design thinking empathy map, and final scope boundary definition.	Completed
M2	Project Planning & Architecture	Milestone scheduling, architecture diagramming, data model schema design, ERD mapping across 5 scoped tables, and UI layout wireframing.	Completed
M3	Data Model & Security Construction	Table creation in scope x_2065601_hospit_0, reference fields, Choice fields, auto-numbering, and ACL security role configuration.	Completed
M4	Business Logic & Workflow Engine	Client Scripts, Business Rules (waitlist auto-promotion logic), Script Includes (atomic slot booking engine), and event triggers.	Completed
M5	Service Portal Development	AngularJS single-page portal widget (hospital_booking), multi-tab role views, server/client script controllers, and CSS styling.	Completed
M6	Testing, Deployment & Documentation	End-to-end integration testing, sample record creation, capstone folder organization, PDF documentation generation, and GitHub upload.	Completed

3. Technical Architecture & Component Mapping

The solution architecture relies on standard ServiceNow platform capabilities, strictly contained within the custom scope to guarantee data isolation and upgradeability:

- **Data Layer:** 5 Custom Tables (x_2065601_hospit_0_patient , x_2065601_hospit_0_doctor , x_2065601_hospit_0_doctor_slot , x_2065601_hospit_0_appointment , x_2065601_hospit_0_waitlist).
- **Logic Layer:** Server-side Script Includes processing synchronous atomic bookings, state transition Business Rules monitoring cancellations and promoting waitlisted records.

- **Presentation Layer:** Service Portal page embedding custom AngularJS widget `hospital_booking` with responsive Bootstrap 3 grid layout and custom CSS styling.

Key Risk Management Strategy:

To mitigate race conditions during simultaneous patient slot bookings, logic is encapsulated in server-side Script Includes using synchronous transaction execution, ensuring slot availability is validated and locked prior to appointment creation.